



Amazon ML Summer School 2025

Sample Test by Scaler

Section 1: MCQ Section

Question 1: Which of the following conditions must be met for the Central Limit Theorem (CLT) to apply?

- A) The sample must be drawn from a normal distribution
- B) The population must be symmetric
- C) The population must have finite mean and variance
- D) The sample size must be at least 30 for any population

Answer: C — CLT holds when population has finite mean and variance.

Question 2: Let $f(x) = \frac{x^3}{e^x}$. What is the value of $\lim_{x \rightarrow \infty} f(x)$?

- A) 1
- B) ∞
- C) Does not exist
- D) 0

Answer: D — exponential in denominator grows faster than cubic numerator.

Question 3: Which of the following is true about dropout in neural networks?

- A) Dropout amplifies gradients
- B) Dropout reduces overfitting by deactivating neurons during training
- C) Dropout increases model complexity
- D) Dropout is applied only during inference

Answer: B — dropout helps regularize the model during training.

Question 4: Which test compares means with unequal variance?

- A) Paired t-test
- B) Welch's t-test
- C) Chi-square test
- D) Z-test

Answer: B — Welch's t-test handles unequal variances.

Question 5: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Which of the following is the inverse of A ?

- A) $\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$
- B) $\begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$
- C) $\begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix}$
- D) $\begin{bmatrix} 2 & -1 \\ -1.5 & 0.5 \end{bmatrix}$

Answer: B — inverse of A is $\frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$.



Question 6: A box contains 3 blue, 4 green, and 5 yellow balls. Three balls are drawn at random without replacement. What is the probability that at least two balls are of the same color?

- A) $\frac{8}{11}$ B) $\frac{3}{11}$ C) $\frac{88}{220}$ D) $\frac{30}{220}$

Answer: A — total combinations: $\binom{12}{3} = 220$; favorable outcomes = 160 (at least 2 same color), so probability = $\frac{160}{220} = \frac{8}{11}$.

Question 7: Which is not an assumption of Linear Regression?

- A) Multicollinearity exists
B) Linearity
C) Normally distributed residuals
D) Homoscedasticity

Answer: A — multicollinearity violates regression assumptions; it is not required, rather avoided.

Question 8: If $\text{rank}(A) = 2$ and A is a 3×3 matrix, which of the following must be true?

- A) $\det(A) = 0$
B) Columns of A are linearly independent
C) Nullity of A is 0
D) A is invertible

Answer: A — rank less than 3 implies $\det(A) = 0$ and A is not full-rank.

Question 9: Two datasets each have standard deviation = 5. If you combine them into a single dataset of double the size, with equal means, what is the combined standard deviation?

- A) 5
B) 2.5
C) 0
D) 10

Answer: A — if means are equal and sizes equal, variance remains same, so standard deviation is unchanged: 5.

Question 10: Which of the following causes Random Forest to overfit?

- A) Fewer features
B) More trees
C) Bootstrapping
D) Deeper trees

Answer: D — deeper trees increase model complexity and risk of overfitting.

Question 11: In F_β score, what happens when $\beta \rightarrow \infty$?

- A) Recall emphasized
B) F-score vanishes
C) Equal weight
D) Precision emphasized

Answer: A — as $\beta \rightarrow \infty$, recall dominates the score.

Question 12: A bag contains 3 red and 2 blue balls. Two balls are drawn without replacement. What is the probability that they are of different colors?

- A) $\frac{4}{5}$ B) $\frac{3}{5}$ C) $\frac{7}{10}$ D) $\frac{2}{5}$

Answer: B — total pairs = $\binom{5}{2} = 10$, different color pairs = 6, so probability = $\frac{6}{10} = \frac{3}{5}$.



Question 13: In gradient descent, what happens if the learning rate is too high?

- A) Training is guaranteed to succeed
- B) Model underfits
- C) Convergence is faster
- D) The model may diverge

Answer: D — large learning rate can cause updates to overshoot, preventing convergence.

Question 14: Let A^{19} . Find the eigenvalues of A^{19} .

- A) $2^{19}, (-2)^{19}$
- B) 1, -1
- C) 2, -2
- D) $2^{19}, -2^{19}$

Answer: D — eigenvalues of A^n are powers of eigenvalues of A .

Question 15: Two fair dice are rolled. What is the probability that the sum is divisible by 4?

- A) $\frac{1}{3}$ B) $\frac{5}{18}$ C) $\frac{1}{6}$ D) $\frac{1}{4}$

Answer: D — favorable sums: 4, 8, 12 $\rightarrow 3+5+1 = 9$ outcomes out of 36, so $\frac{9}{36} = \frac{1}{4}$.

Question 16: Which model is most sensitive to outliers?

- A) Naive Bayes
- B) Decision Tree
- C) KNN
- D) Logistic Regression

Answer: C — KNN bases prediction on nearby points, so outliers distort distance-based decisions.

Question 17: Let $f(x) = x^2 \sin x$. What is $f''(x)$?

- A) $x^2 \cos x - 4x \sin x + 2 \cos x$
- B) $2 \cos x - x^2 \sin x$
- C) $2 \sin x + 4x \cos x - x^2 \sin x$
- D) $2 \sin x + x^2 \cos x$

Answer: C — $f'(x) = 2x \sin x + x^2 \cos x$, then $f''(x) = 2 \sin x + 2x \cos x + 2x \cos x - x^2 \sin x = 2 \sin x + 4x \cos x - x^2 \sin x$.

Question 18: A binary classifier shows the following results on a test set of 1000 samples:

True Positives (TP) = 200, False Positives (FP) = 50, False Negatives (FN) = 100, True Negatives (TN) = 650

What is the F1-score?

- A) 0.73
- B) 0.80
- C) 0.60
- D) 0.67

Answer: A — $F1 = \frac{2 \cdot 0.8 \cdot 0.67}{0.8 + 0.67} \approx 0.73$

Question 19: In XGBoost, what does the regularization term in the objective function help prevent?

- A) Gradient vanishing
- B) Early stopping
- C) Overfitting
- D) Underfitting

Answer: C — regularization penalizes complexity to reduce overfitting.



Question 20: Given: $\text{Var}(X) = 1$, $\text{Var}(Y) = 4$, $\text{Var}(2X - 3Y) = 34$. Find the correlation coefficient ρ_{XY} .

- A) 0
- B) 0.5
- C) 0.25
- D) -0.5

Answer: C — Use identity: $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$.
So, $34 = 4 + 36 + 2(2)(-3)\rho\sqrt{1 \cdot 4} \Rightarrow \rho = 0.25$

Section 2: DSA Type Coding Section

TEST INSTRUCTIONS

- **GENERAL**

- This is a timed test. Please keep track of the time remaining at the top of the page.
- If you feel the question is unclear, try out different inputs and check the expected output for each of them.

- **CODE IMPLEMENTATION**

- You only need to implement the given function.
- Do not read input; instead use the arguments to the function.
- Do not print the output; instead return values as specified.
- Still have a question? Check out Sample Codes for more details.

- **PLAGIARISM**

- Your code submission will be checked for plagiarism after you're finished.
 - If the code is found to be similar to another, it will be reported as appropriate.
 - To prevent plagiarism, we discourage switching tabs or windows during the test. Any such activity will be recorded and will be available for the company contact to see.
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Q1. Streetlight Motion Detection Duration

Problem Description

In a city, a streetlight turns on automatically and stays on for exactly B seconds each time it detects motion. If it detects motion again before turning off, the timer resets, and it stays on for another B seconds from the new detection time.



You have a list A where each element $A[i]$ indicates the time when the streetlight detects motion, and an integer B . Calculate the total time the streetlight remains on.

Problem Constraints

$1 \leq A.length \leq 10^5$ $0 \leq A[i], B \leq 10^9$ A is sorted in non-decreasing order.

Input Format

First Argument is an Integer Array, A denoting the time when the streetlight detects motion.
Second Argument is an Integer, B denoting the duration in seconds.

Output Format

Return an Integer, denoting the total time the streetlight remains on.

Example Input

Input 1: $A = [1, 4], B = 2$

Input 2: $A = [1, 2], B = 2$

Example Output

Output 1: 4

Output 2: 3

Example Explanation

Explanation 1: At second 1, the streetlight detects motion and turns on for seconds 1 and 2. At second 4, the streetlight detects motion again and turns on for seconds 4 and 5. The streetlight is on for seconds 1, 2, 4, and 5, which is 4 seconds in total.

Explanation 2: At second 1, the streetlight detects motion and turns on for seconds 1 and 2. At second 2, the streetlight detects motion again and resets the timer. It stays on for seconds 2 and 3. The streetlight is on for seconds 1, 2, and 3, which is 3 seconds in total.

Solution:

```
1  int Solution::findTotalTime(vector<int> &A, int B) {
2      long long totalTime = 0;
3      long long lastOnTime = -1;
4
5      for (int time : A) {
6          if (time > lastOnTime) {
7              totalTime += B;
8          } else {
9              totalTime += time + B - lastOnTime;
10             }
11             lastOnTime = time + B;
12         }
13
14     return totalTime;
15 }
```



Q2. Recover Binary Search Tree

Problem Description

Two elements of a Binary Search Tree (BST), represented by root A , are swapped by mistake. Tell us the 2 values, when swapped, will restore the Binary Search Tree (BST).

A solution using $O(n)$ space is pretty straightforward. Could you devise a constant space solution?

Note: The 2 values must be returned in ascending order.

Problem Constraints

$1 \leq \text{size of tree} \leq 100000$

Input Format

First and only argument is the head of the tree, A .

Output Format

Return the 2 elements which need to be swapped.

Example Input

Input 1: A binary search tree where the root is 1, its left child is 2, and its right child is 3.

Input 2: A binary search tree where the root is 2, its left child is 3, and its right child is 1.

Example Output

Output 1: [2, 1]

Output 2: [3, 1]

Example Explanation

Explanation 1: Swapping 1 and 2 will change the BST to be a valid BST where the root is 2, its left child is 1, and its right child is 3.

Explanation 2: Swapping 1 and 3 will change the BST to be a valid BST where the root is 2, its left child is 1, and its right child is 3.



Solution:

```
1  vector<int> Solution::recoverTree(TreeNode* A) {
2      TreeNode *curr = A, *prev = nullptr;
3      TreeNode *first = nullptr, *second = nullptr;
4
5      while (curr) {
6          if (!curr->left) {
7              if (prev && prev->val > curr->val) {
8                  if (!first) first = prev;
9                  second = curr;
10             }
11             prev = curr;
12             curr = curr->right;
13         } else {
14             TreeNode *temp = curr->left;
15             while (temp->right && temp->right != curr) {
16                 temp = temp->right;
17             }
18             if (!temp->right) {
19                 temp->right = curr;
20                 curr = curr->left;
21             } else {
22                 temp->right = nullptr;
23                 if (prev && prev->val > curr->val) {
24                     if (!first) first = prev;
25                     second = curr;
26                 }
27                 prev = curr;
28                 curr = curr->right;
29             }
30         }
31     }
32
33     vector<int> result = {first->val, second->val};
34     sort(result.begin(), result.end());
35     return result;
36 }
```

Important Note

These questions are provided by Scaler for practice for the ML School 2025 Selection Test. You will receive a sample test invitation link after applying and submitting the required form.

For more content and guidance, visit my GitHub: <https://github.com/cu-sanjay>
If you find it helpful, please star the repository and follow the account.

All rights are reserved by **Amazon** and **Scaler (InterviewBit)**.
Thank you. I wish you the best for the ML Summer School exam and program.

